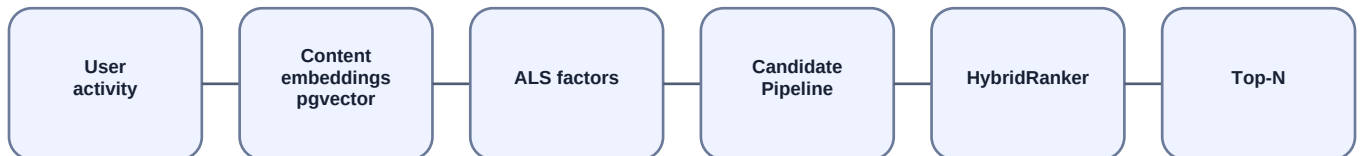


WatchMan AI Hybrid Recommendation & Automation

Content embeddings + ALS + hybrid ranking + Celery

Recommendation architecture

The live design combines semantic content similarity and collaborative filtering, then applies popularity/freshness/preference signals through the existing HybridRanker.



Content-based branch

The existing Hugging Face **all-MiniLM-L6-v2** path produces 384-dimensional vectors. Content vectors are stored in PostgreSQL/pgvector; user taste vectors are derived from weighted interacted-item vectors and normalized.

Collaborative branch: ALS

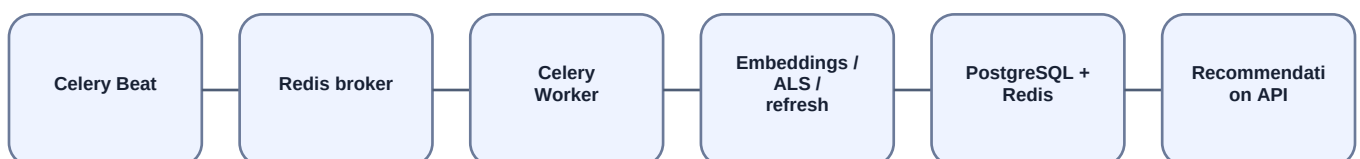
ALS builds a user-item interaction matrix from signals such as ratings, saves, watches and events. The audited implementation uses 64 latent factors and persists versioned user/item factors.

Hybrid signals

Signal	Purpose
Content	Semantic similarity to learned taste.
ALS collaborative	Learns cross-user/cross-item patterns.
Popularity	Broad relevance and cold-start support.
Freshness	Recency signal.
Preferences	Explicit taste signal.
Diversity	Limits over-concentration.

Audited default weights: content 0.40, collaborative 0.30, popularity 0.15, freshness 0.10, preference 0.05.

Automation workflow



Background jobs handle missing embeddings, ALS training and recommendation refresh/persistence. The normal recommendation API should not train ALS or embed the whole catalog synchronously.

No K-Means/clustering is part of the current recommendation design.